

RICCARDO LODDO

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WORK EXPERIENCE

Data Engineering Intern – Advancia Technology, Viareggio (LU), Italy

June 2025 - Present

Hands-on experience with SQL, Python, and ETL processes

- Developed SQL queries and Python scripts for data extraction, transformation, and analysis.
- Worked on real ETL processes, understanding workflow and data pipeline management.
- Completed a Data Engineering course, gaining practical knowledge of Hadoop, Spark, Hive, Kafka, and related big data technologies.
- Familiarity with RDBMS and NoSQL databases, and experience in managing large datasets.

Professor of Physics – Liceo Gandini, Lodi, Italy

Dec 2024- June 2025

Teaching Mathematics and Physics to High School Students

- Designing lesson plans and assessments for mathematics and physics courses.
- Providing personalized support to improve students' understanding and performance.

Professor of Mathematics and Physics – ITT Artemisia Gentileschi, Milano, Italy Oct 2024 - Dec 2024

Teaching Mathematics and Physics to High School Students

- Designing lesson plans and assessments for mathematics and physics courses.
- Providing personalized support to improve students' understanding and performance.
- Got an AI course for teachers with the aim of teaching AI to students in the future.

Master's Thesis – Università degli Studi di Milano, Italy

Dec 2023 - Spt 2024

Orbital Reconstruction and Disc Morphology: A Study of MWC 758 Following a Flyby Event Supervisor: Giuseppe Lodato

- Conducted research on the morphology and kinematics of circumstellar discs, focusing on accretion discs and their structure and dynamics.
- Developed a theoretical model to explain the observed spiral structures and ring concavities, investigating theories such as fragmentation, planetary formation, and flyby events.
- Focused on the role of flyby events in shaping disc morphology, utilizing cinematic gas data from ALMA (Atacama Large Millimeter/submillimeter Array) in Chile.
- Skills: Phantom, Splash, Python

Bachelor's Thesis – Università degli Studi di Milano, Italy

Feb 2022 - Oct 2022

Search for ultra-high energy photons in data of the Pierre Auger Observatory. Supervisor: Lorenzo Caccianiga

- Conducted research on ultra-high-energy cosmic rays (UHECRs) and studied potential anisotropies in the distribution of high-energy photons.
- Collaborated with the Pierre Auger Observatory to investigate source catalogs, including neutron stars, magnetars, pulsars, and microquasars.
- Performed spatial correlation analysis between observed photon events and candidate sources, and conducted an autocorrelation analysis to detect anisotropies with low statistics.
- Skills: Python

EDUCATION

Università degli Studi di Milano

Oct 2022 – Spt 2024

Master of Astrophysics, Cosmology and Space Science

Milan, Italy

- Final grade: 110/110
- Focus: Circumstellar and Protoplanetary Discs, Flyby Events, Accretion Disc Morphology and Dynamics
- Relevant Coursework: Computational Astrophysics, Advanced Cosmology, Planet Formation, High-Energy Astrophysics

Universidad Autónoma de Madrid - UAM

Spt 2023 – March 2024

Erasmus program scholarship

Madrid, Spain

- Focus: During the Erasmus program in Spain, I pursued theoretical physics in depth. I worked in teams with international students, adapting to new methodologies and a challenging academic environment.

- Relevant Coursework: Theoretical Physics, Cosmology, Advanced mathematics, Lie groups.

Università degli Studi di Milano

Oct 2018 – Oct 2022

Bachelor of Physics

Milan, Italy

- Final grade: 97/110
- Focus: High-Energy Cosmic Rays, Astrophysics of High-Energy Particles, Anisotropies in Photon Distribution
- Relevant Coursework: Cosmic Rays, Quantum Physics, Laboratory of Astronomy, Relativity, Electromagnetism

Istituto Istruzione Superiore Alessandro Volta

Spt 2013 – June 2018

Specialization in mechanics, mechatronics and renewable energy

Lodi, Italy

- Final grade: 90/100
- Focus: I gained practical knowledge in thermal systems (boilers), city-wide infrastructure, and landscape design. Completed an additional 10-day renewable energy course in Padua in collaboration with the University of Padua, focused on sustainable energy solutions. Experienced in 2D and 3D technical design using AutoCAD and other CAD tools.

TECHNICAL PROJECTS

End-to-End ETL Process Implementation

June 2025 - Present

- Designed and implemented an end-to-end ETL process using SQL and PL/SQL, handling data ingestion from multiple sources into a staging area and loading it into a data warehouse.
- Applied data filtering, transformation, and selection techniques using stored procedures and SQL scripts, ensuring data quality and integrity.
- Managed error handling, logging, and bad data tracking to guarantee reliability and traceability of the pipeline.
- Developed a procedure to dynamically select the optimal method for populating external tables (External Tables, UTL_FILE, and alternative approaches).
- Gained practical experience in pipeline orchestration, data management, and end-to-end process execution in a real-world data engineering environment.

Course: Machine Learning

Gen 2025 – March 2025

- Explored fundamental concepts of machine learning, including classification, regression, and clustering techniques.
- Implemented various ML algorithms such as K-Nearest Neighbors, Naive Bayes, Support Vector Machines, and Neural Networks.
- Developed hands-on experience using TensorFlow for building and training neural networks for classification and regression tasks.
- Applied dimensionality reduction techniques like Principal Component Analysis (PCA) to optimize feature selection.
- Worked with Python and Google Colab to preprocess data, train models, and evaluate performance using appropriate metrics.

Course: Statistical and Computational Techniques in Astrophysics

Oct 2023 - Dec 2023

- Developed statistical and computational skills for astrophysical data analysis, focusing on Bayesian inference and Python programming. Knowledge of Monte Carlo methods
- Analyzed astronomical data from major archives to formulate models and solve astrophysical problems.
- Applied statistical methods, including clustering algorithms and Monte Carlo techniques, on projects like stellar clusters from Gaia.

Course: Data Analysis Methodologies

Oct 2022 - Apr 2023

- Introduced to statistical analysis of experimental data, focusing on C++ programming in ROOT environment.
- Applied Monte Carlo simulation methods, maximum likelihood estimation, and hypothesis testing to analyze data samples.
- Developed quantitative skills in data classification, error propagation, and statistical inference, enhancing problem-solving capabilities.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

Tools & Frameworks: Phantom, Splash, Git, ALMA Data Reduction, Monte Carlo methods, Markov chain.

Technical Areas: Algorithms, Computational Astrophysics, Data Analysis, High-Energy Physics, Numerical Methods

LANGUAGES

Italian (Mother tongue), English (Upper Intermediate), Spanish (Basic user)